



# Cybersecurity and AI Case Studies

## Lab Logbook

SID:2467121

**MOD 006570**

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LAB Logbook github link: <https://github.com/Erick-254/Cyber-security-and-AI-case-studies>

## Lab 1

1. Write one sentence each for any of the five Pandas classes you find interesting.

Data Frame – A two-dimensional, size-mutable, and heterogeneous tabular data structure with labeled rows and columns.

Examples: Create a sample table of network traffic.

Series – A one-dimensional labeled array capable of holding any data type, similar to a column in a spreadsheet.

Example: Create a single column of packet lengths.

Categorical – A data type for storing categorical variables efficiently, often used for repeated string values.

Example: Store attack types efficiently

Datetime Index – An index specifically designed to handle datetime objects for time-series data analysis.

Example: Use timestamps as an index for time-series analysis.

Group By – A class that allows you to split a Data Frame into groups based on some criteria and apply aggregation or transformation functions.

Example : Group attacks by type and count them:

## Lab 2

1. SolarWinds Supply Chain Attack (2020) – Attackers inserted a backdoor into

SolarWinds' Orion software, impacting thousands of organizations.

2.

Useful

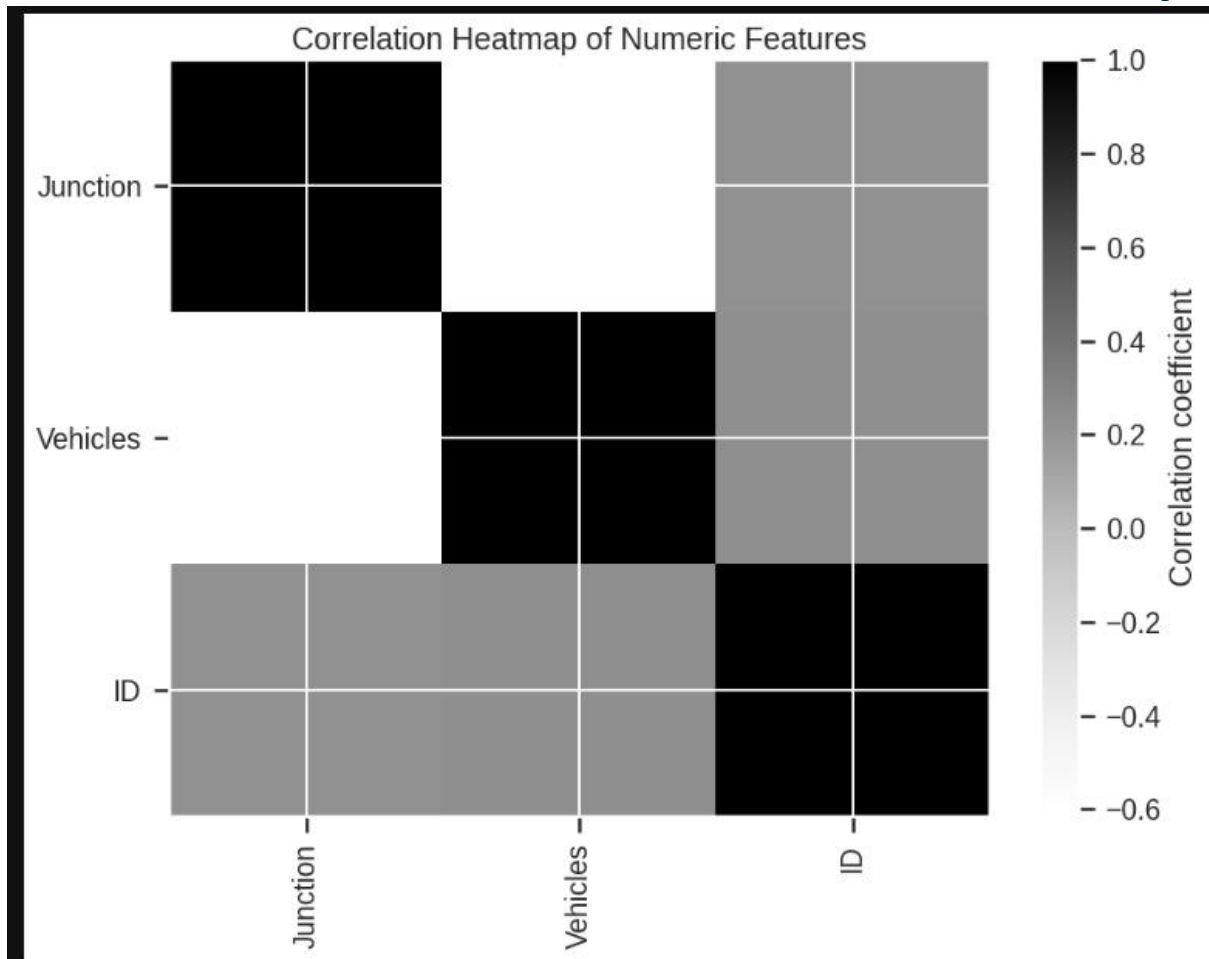
Reference:

[SolarWinds Cyberattack – Wikipedia](#)

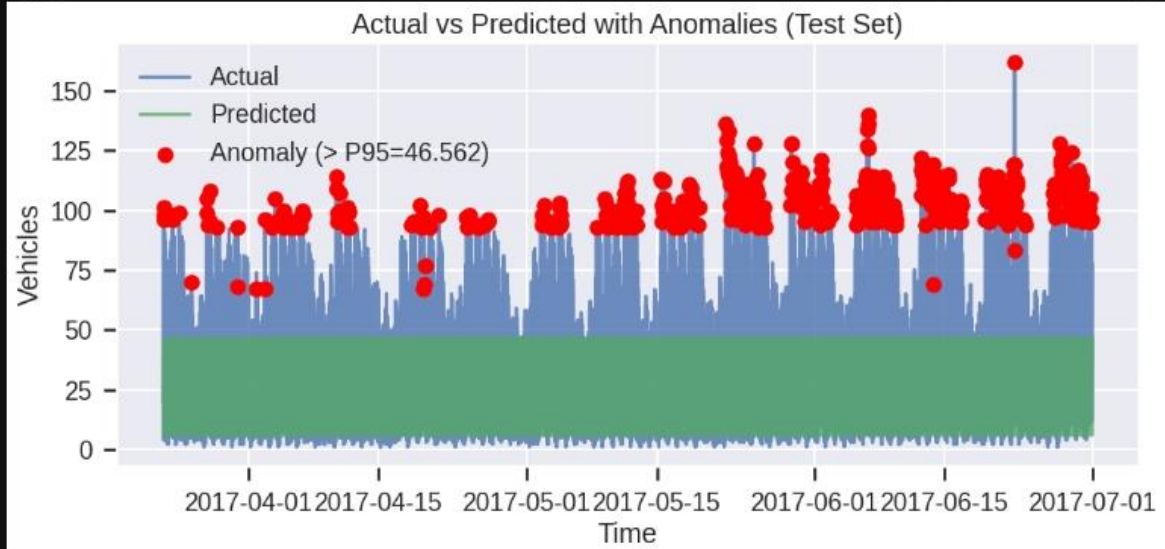
### Lab 3

EDA

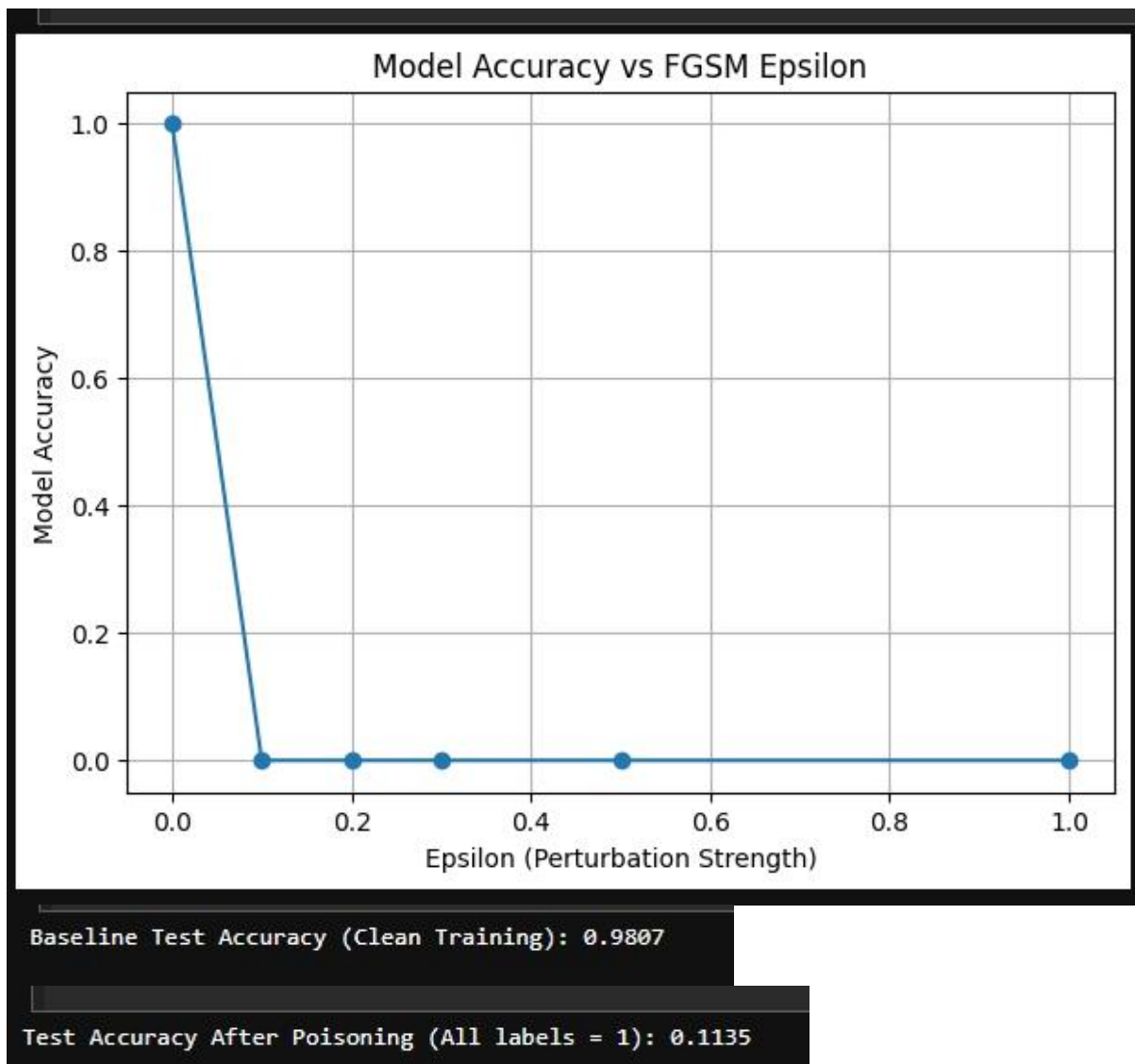
plot



```
Out[ ]:      threshold      value  anomaly_count  anomaly_frac
2  P90=31.576  31.576190         962         0.099990
0  P95=46.562  46.562360         481         0.049995
1  P99=64.478  64.477882          97         0.010082
```

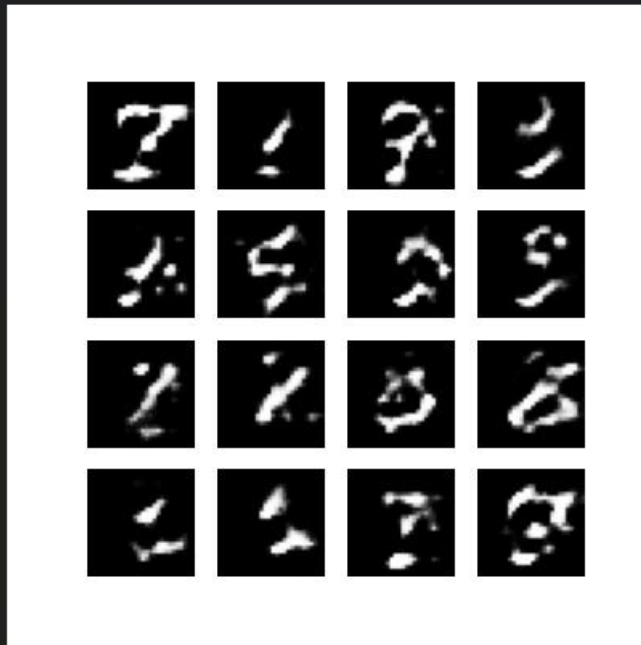


Lab 4

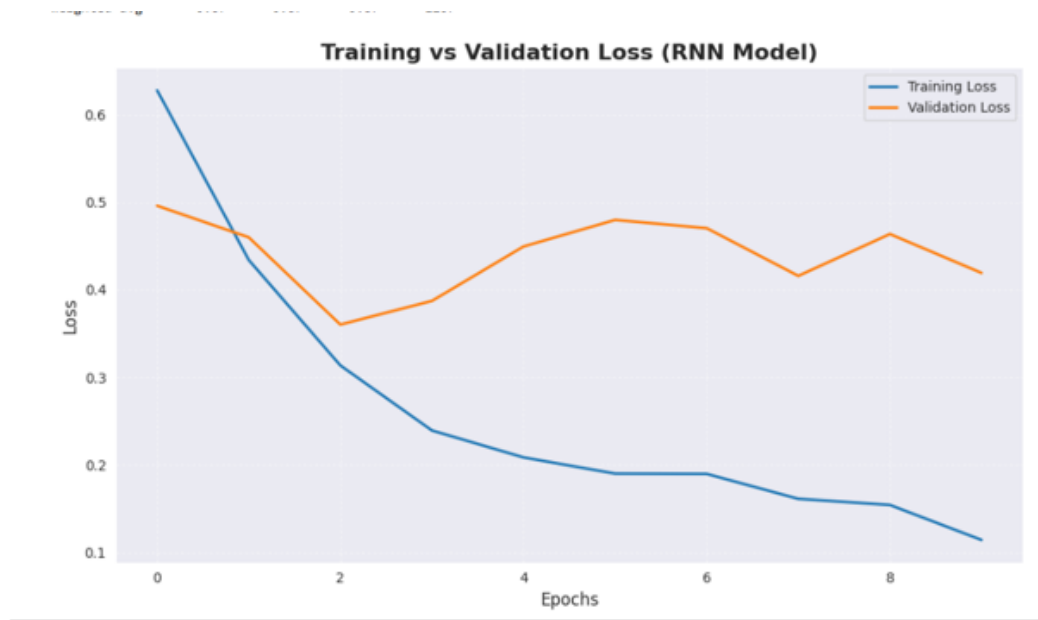


## Lab 5

image\_at\_epoch\_0021.png



Lab 6



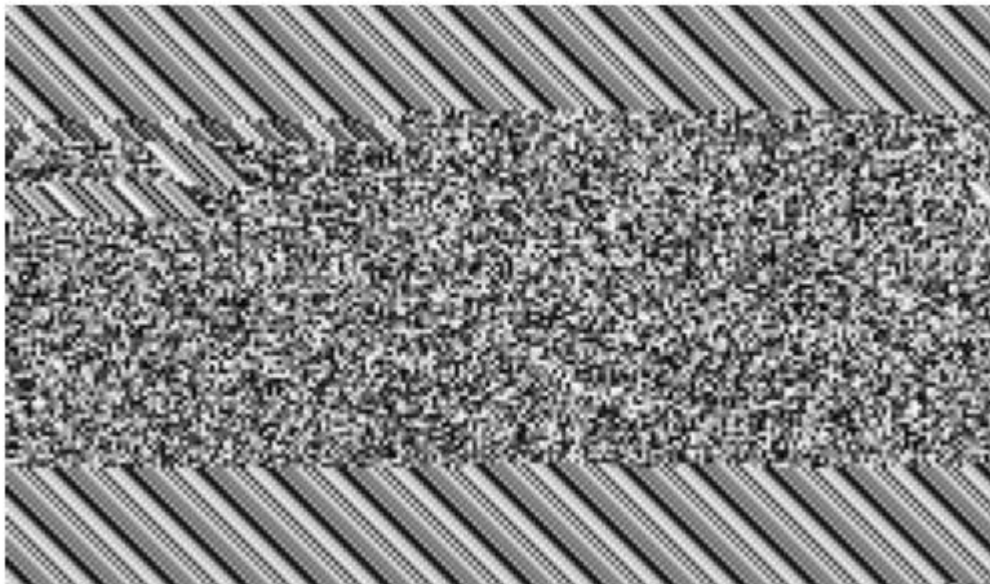
## Lab 7

Real image



Cypher image

Encrypted image saved as encrypted\_ecb.png



## Lab 8

Mark Rutto

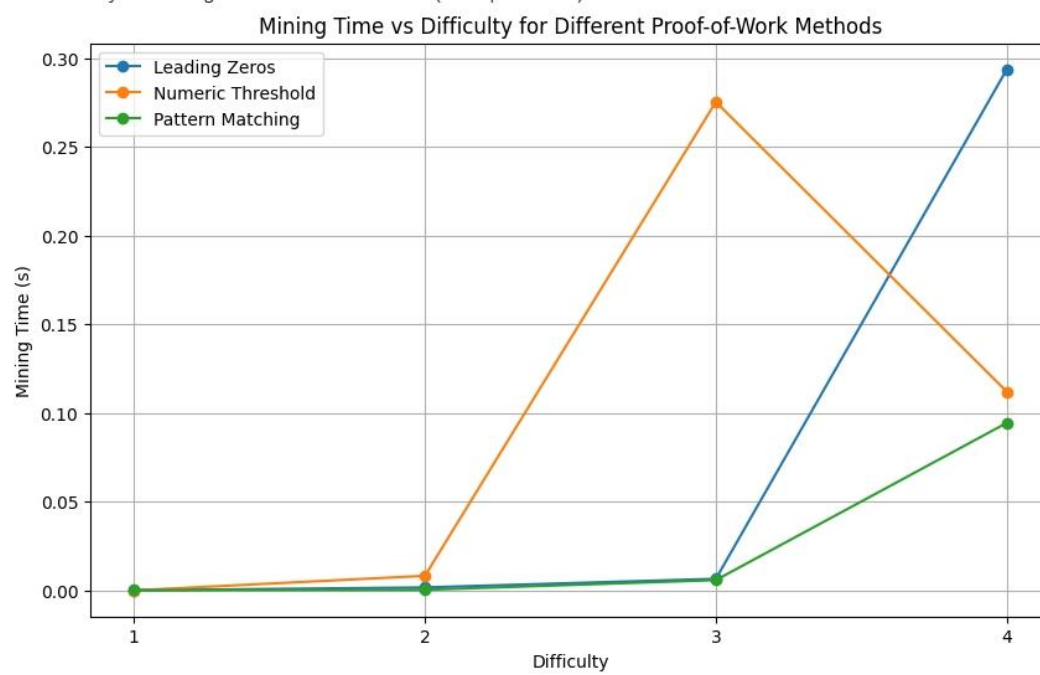
P 23

G 5

S 18

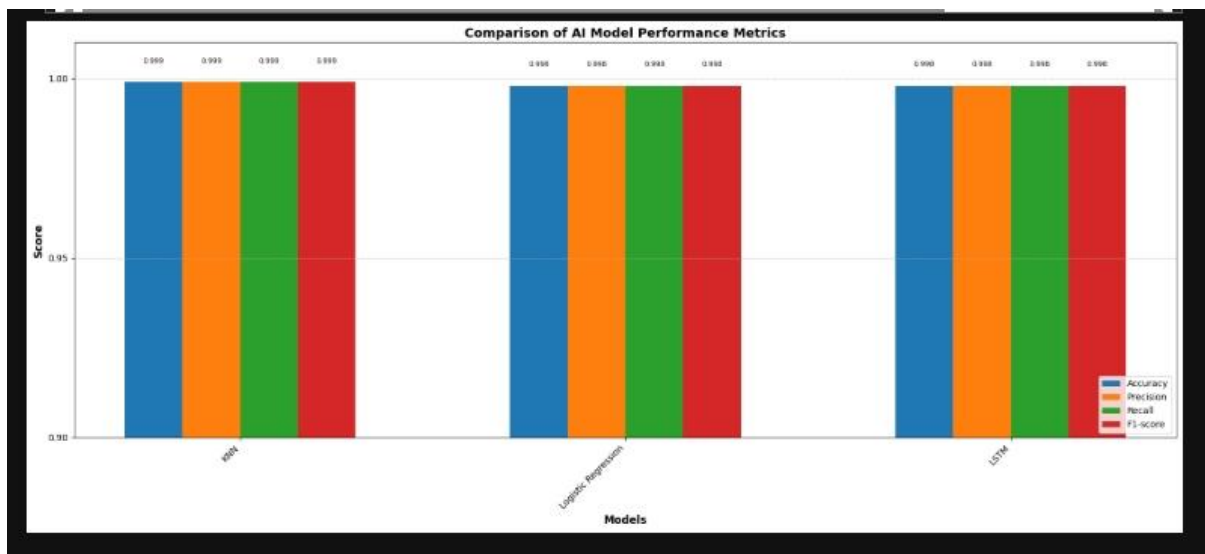
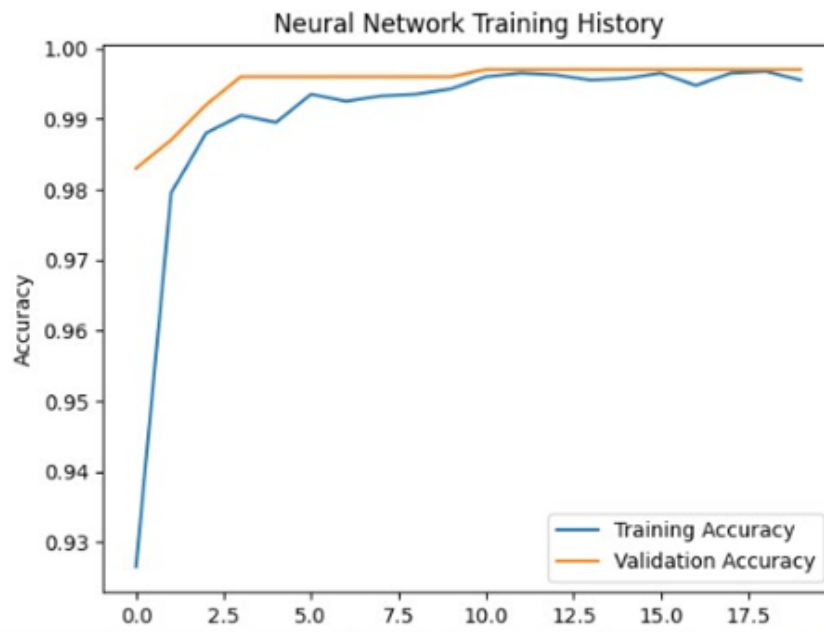
Private key= 16

## Lab 9

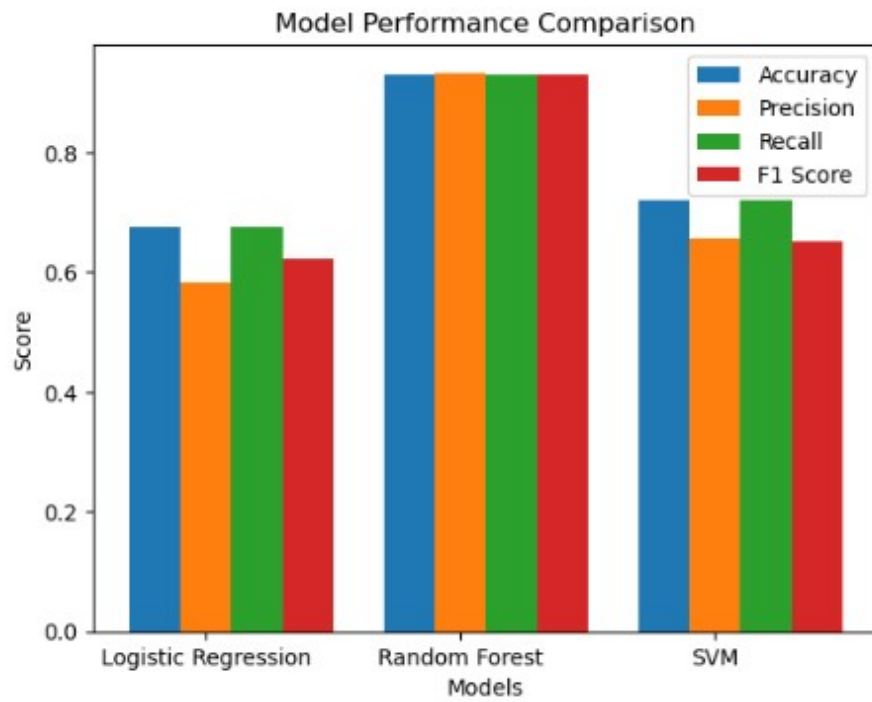


## Lab 10





Lab 11



## Lab 12

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